PIP

```
initialism for Preferred Installer Program
used to install different python packages
```

Packages

```
are modules that contain one or more modules or other packages
```

Install PIP

```
1. Go to terminal or command prompt
2. type pip install pip
3. check if installed, pip --version
```

Install Packages by PIP

| numpy

```
abbreviation for numeric python
popular for machine learning and data science
fundamental for scientific computing
```

> **Important**
>
> details in obsidian

```
np_arr = numpy.array(list)
# similar to lists but more memory efficient, faster, and designed for
numerical computation
```

| pandas

```
open source, BSD-licensed library (means you can use it freely, even selling
it)
easy to use data structures and data analysis tools
```

```
pip install pandas
import panas
```

| webbrowser

```
to open any website
installed by default in python3
```

```
import webbrowser
url = ""
webbrowser.open_new_tab(url)
```

> **Important**
>
> more in obsidian

| beautifulsoup4 library

```
turn row HTML to structured object
cooks the HTMl, separate the tags
```

```
html = response.text
soup = BeautifulSoup(html, "html.parser")

#find tags
soup.find("table") #the first table
soup.find_all("a") #all a tags

#get the text
soup.title.text
```

Uninstalling Packages

```
pip uninstall package_name
```

List of Packages

```
to see installed packages on your machine
```

```
pip list
```

Show Package

```
to show information about a package
```

```
pip show packagename
pip show --verbase packagename #for more details
```

PIP Freeze

```
to generate installed packages with their versions, output is suitable for
requirement file
```

```
pip freeze
```

Reading from URL / API

| API

```
initialism for Application Program Interface
an exchange structured data between servers,, allows opening a netweork and
implement CRUD
```

| Install request

```
pip install requests
```

```
response = requests.get(url) #open a network and fetch data from url
response.status_code
response.headers #metadata how the data is interpreted
response.text #extract text from fetched response object
reponse.json() #extract json data
```

| Read from API

```
data = response.json()
```

Creating a Package

```
-folder
    |- __init__.py
    #python recognize this as a package, expose specified resources to be
imported, make all functions available
```

```
    |- file.py
    |-file2.py
```